AirTrust study and simulation guide

Team 10 • Project 9 • Design of Smart Cities • AY 2026–27

Use pages 1–4 to study the project, pages 5–7 as the live demonstration script, and page 8 for viva preparation. The demonstration takes about 10 minutes. All values describe the implemented historical-data prototype.

The project in one sentence

Say "We built AirTrust to predict and map Delhi's PM2.5 using federated learning across eight simulated geographic clients, while testing how sensor quality affects the shared model."

A 45 second opening for ma'am

Say "Ma'am, our assigned topic is Federated Learning for City-Wide Air Pollution Mapping Without Centralizing Sensor Data. We use real Delhi reference-monitor observations to generate controlled virtual low-cost sensor errors. Local clients train models and send model parameters to an aggregator. We compare trust weighting with FedAvg, centralized learning, kriging and robust methods. We evaluate both future times and locations excluded from training. Our key finding is that early input correction improves this simulation, while trust weighting alone does not beat FedAvg consistently."

What happens from input to output

1. Download public observations and station coordinates; retain valid, sufficiently complete stations.
2. Group 36 locations into eight geographic clients and simulate noisy readings.
3. Estimate calibration confidence from early co-located virtual/reference pairs.
4. Train local graph or temporal models, aggregate parameters, and select by validation error.
5. Score saved models against reference truth, then display results and historical maps.

Terms you must understand

PM2.5: fine particles at most 2.5 micrometres in diameter. Concentration is measured in μg/m³, not AQI. Client: one geographic group of stations, not one person or one station. Federation: local optimization followed by combining model parameters. Non-IID: clients have different pollution distributions, coverage and simulated sensor errors.

Calibration: relating sensor readings to a reference. Co-location: comparing readings at the same location. Drift: error changing over time. Held-out station: a location whose readings are excluded from fitting, validation, calibration and donor histories in that evaluation.

The boundary of our claim

This is a single-computer simulation with public data. The aggregation function receives parameters and scalar weights. The simulator and evaluator can load the public raw observations. We have not deployed independent physical sensors, implemented cryptographic privacy, or measured health improvements. SDG links are exposure monitoring (3), urban air quality (11.6) and environmental observation (13).

Datasets and database answers

Exactly what we used

Dataset and period	Source rows	Role in this project
Delhi PM2.5, hourly, 2024	317,627	Primary reference labels for FL simulation
Delhi PM10, hourly, 2024	323,061	Contextual pollutant exploration
Delhi NO_2, hourly, 2024	325,379	Contextual pollutant exploration
Beijing, Mar 2013–Feb 2017	420,768 station-hours	Independent PM2.5 persistence benchmark

Delhi is from India Air Quality Database, XKDR Forum, compiling CPCB's Continuous Ambient Air Quality Monitoring network (CAAQM). Its 39 catalog entries supply coordinates and hourly records. After cleaning/coverage/co-location checks, the PM2.5 experiment retains 36 distinct sites and 300,355 valid observations over an 8,784-hour 2024 grid.

Beijing is Song Chen's UCI Multi-Site Air Quality dataset, with 12 reference stations. The source includes multiple pollutants and meteorological variables; our secondary benchmark uses PM2.5. It does not test Beijing FL or Delhi-to-Beijing transfer. Delhi PM10 and NO_2 are also not extra PM2.5 training labels.

Counts to remember: Delhi totals 966,067 pollutant rows; 648,440 were added. Including Beijing gives 1,386,835 source rows, but the schemas differ: Delhi rows contain one pollutant, Beijing rows contain several variables. Weekly copies, nested archives and synthetic derivatives are not additional independent data.

If ma'am asks which database we built

Say "Our application uses a file-based data pipeline rather than a SQL database. Raw observations are CSV files, station metadata and audits are JSON, prepared arrays are NPZ, and trained PyTorch checkpoints are PT files. Pandas loads and joins these files, and FastAPI exposes the results to the dashboard. The India Air Quality Database is our external data provider; it is not a database server we built."

Delhi CSV columns: station_id, parameter_name, unit, period_start, mean, min, max, n. Here mean is the hourly aggregate and n is its contributing observation count. Join station_id to station metadata for station_name, latitude and longitude. The logical observation key is station + pollutant + timestamp. Delhi times are local IST; Beijing times remain local to Beijing.

Cleaning and licensing

PM2.5 values below 0 or above 2,000 µg/m³ are excluded. Stations require at least 70% valid hourly coverage. Catalog entries within 50 m are consolidated by completeness. Input gaps can be forward-filled for at most three hours; reference labels are never filled. Features are scaled by fixed constants, not fitted on test data.

Both source datasets are CC BY 4.0: redistribution needs attribution and disclosure of modifications. Our bundle includes sources and checksums. Reference monitors are real; low-cost errors are synthetic. OpenAQ and sensor.community were proposal candidates but contribute no data to the reported results.

Sources and evidence: github.com/xKDR/Air-Quality-Database; doi.org/10.24432/C5RK5G; sensor study doi.org/10.5194/amt-11-4823-2018. Exact counts and scores are in datasets/manifest.json, results/spatial_cv/summary.csv and results/calibration/summary.csv; extended interpretation is in docs/EXTENDED_EVALUATION.md.

Models and federated learning

Why use two kinds of model

The spatial graph neural network (GNN) predicts a target using up to three nearby seen stations within its client. Each donor supplies six strictly past PM2.5 readings plus relative coordinates. Shared message layers and attention pool the donor information. The target's own reading is not an input, allowing prediction at an unseen location.

The long short-term memory network (LSTM) learns temporal patterns from a known target station's own six-hour history. It is a separate forecasting task. Its smaller error cannot be used to claim better unseen-location mapping because it receives information the spatial model does not.

One training round explained

The server broadcasts current parameters. Each of eight clients starts with that model, trains on its local noisy samples and returns updated parameters. The server combines updates and broadcasts the new model. We use 20 global rounds and 12 local optimizer steps per client per round. Twelve steps means 12 minibatch updates, not 12 complete epochs.

Settings: Adam learning rate 0.002; minibatch up to 128; gradient clipping norm 5; seeds 17, 29 and 43. A seed controls random simulation/training choices, not a new dataset. The best checkpoint is selected by clean known-station validation RMSE; test scores never select it. Clean-reference validation access is an assumption of this benchmark.

How aggregation methods differ

Method	What it does
FedAvg	Weights each client model by its number of training samples.
Trust weighting	Uses sample count × calibration confidence, normalized across clients.
Equal / median / trimmed mean	Equal average; coordinate median; or discard one extreme value per side.
Krum	Selects the update most consistent with neighbours; configured f = 1.
Consistency / temporal reputation	Weights update agreement instantly or smooths agreement across rounds.
Centralized / IDW / kriging	Pooled neural training; inverse-distance interpolation; covariance-based interpolation.

Trust confidence is exp(−calibration RMSE / 35), floored at 0.05. RMSE is measured in the first 10% of hours from virtual sensors paired with their own reference locations. This avoids treating real spatial differences between distant stations as calibration error. The scale 35 is a fixed scenario setting, not a learned hardware specification.

Sensor realism and limitations

The main stress test specifies bias, gain, annual drift, Gaussian noise, dropout and stuck readings; three of eight clients are severely corrupted. Separate sensitivity uses Zheng et al.'s Kanpur relative-error magnitudes (35%/46%) with an explicitly assumed mean-one lognormal generator. It is not a field-fitted Delhi sensor model. Temporal reputation is project-defined. Krum's f = 1 guarantee does not cover three corrupted clients.

Evaluation and results to remember

Two different tests

Chronology: first 10% of hours calibrate, next 60% train, next 15% validate, final 15% test. The future test period starts 7 November 2024 at 02:00 IST. Six-hour histories predict the next target hour; samples are taken every three hours. A training-era reserve is excluded from optimization for the privacy probe.

Original holdout: eight entirely unseen stations, 28 seen stations, repeated over three seeds. Overview and Experiments show this primary benchmark. Extended station CV: five rotating folds × three seeds; every retained station is unseen once per seed. Four folds hold out eight sites and the fifth holds out four. Simulation lab shows these CV results.

What the metrics mean

MAE is average absolute concentration error. RMSE gives greater weight to large errors. Both use µg/m³ and lower is better. R^2 compares squared error with a mean-value baseline; 1 is ideal, 0 matches that baseline, and negative values are possible. "± SD" describes run variability, not a guarantee or a confidence interval.

Unseen-station CV method	MAE ± SD	RMSE ± SD	R^2 mean
FedAvg	68.20 ± 9.47	96.16 ± 12.11	0.466
Trust FL	68.01 ± 8.83	96.69 ± 11.35	0.465
Centralized graph	71.20 ± 9.53	99.25 ± 11.35	0.434
Ordinary kriging	80.84 ± 19.19	109.80 ± 25.11	0.241

These are equally weighted means over 15 fold-seed runs. Folds share underlying data; they are not 15 independent city datasets. The original fixed-holdout trust RMSE is 99.92 ± 7.05, versus FedAvg 98.11 ± 3.98. Do not confuse these with the CV table above.

The strongest supported improvement

A separate input-calibration ablation fits a linear correction per seen station using early co-located pairs, then corrects later noisy inputs and training labels. On the original fixed holdout, trust RMSE falls from 99.92 ± 7.05 to 78.67 ± 2.02 µg/m³ (about 21.3%). Corrected FedAvg reaches 78.49 ± 2.05. This ablation is not a five-fold CV result.

Say "Our experiments show that correcting sensor inputs helps this controlled simulation. They do not establish a consistent benefit from trust weighting over FedAvg. Weighting updates alone cannot remove bias from donor measurements, and down-weighting clients may also lose useful geographic information."

Privacy and communication numbers

Graph trust membership AUC is about 0.502 ± 0.005 for the specified probe, close to random ranking. That is not a privacy guarantee. A graph model has 1,162 parameters; 20 rounds across eight clients account for 743,680 uploaded bytes and the same downloaded bytes, about 1.49 MB total decimal payload. This excludes protocol/calibration overhead and is not measured radio traffic or energy.

Demo part one from launch to comparisons

Target time: 4 minutes. Keep the laptop plugged in. Use a normal browser window large enough to see the sidebar. Do not start full training during the assessed demo; the app uses saved models/results and performs fresh map inference and sensor diagnostics.

Before ma'am arrives

Open the airtrust folder and run start.bat; leave its terminal open. Open http://127.0.0.1:8000/#overview. If the app is already running, simply refresh the page. On a new clone, first install requirements and run python scripts/prepare_local_data.py. Keep the final report, review slides, dataset ZIP and GitHub README ready in separate tabs/windows.

1 Overview about 40 seconds

Do Click Overview in the left sidebar. Point to 36 monitors, 300,355 valid observations and eight clients. Give the opening on page 1.

Say "These are the retained PM2.5 experiment counts. The expanded pollutant-source count is larger. The error and comparison shown here are saved three-seed fixed-holdout results, not live training progress."

2 Pollution map about 70 seconds

Do Click Pollution map or Explore pollution map ↗. Keep the initial test hour. With Model set to Trust-weighted FL, click Update map. Select FedAvg, then click Update map again at the same hour.

Say "We are comparing saved models using the same historical input time. The graph learner predicts a spatial surface from donor histories. Map changes are model predictions, not new ground-truth measurements."

Do Click a hollow station circle, or its station-list button; read whether it is held out. Then move the Test-period hour slider and press Update map. The initial hour 8000 corresponds to 29 November 2024, 08:00 IST. Every model/hour change needs Update map.

Say "Hollow markers identify locations excluded from fitting in the primary holdout. Reference readings shown for evaluation are not their training inputs. The station chart uses daily means. The 28 × 28 grid is a model surface inside a coordinate rectangle; it is not a validated street-level map or AQI classification."

3 Federation about 50 seconds

Do Click Federation. Point to Client calibration & aggregation and Validation convergence. Scroll to Communication accounting.

Say "Clients with larger early co-location errors receive lower confidence. We compare a virtual sensor with its own reference site, avoiding nearest-station gradients. The convergence chart records training already run; the lowest validation error selects the checkpoint. Byte counts are logical parameter payloads."

4 Experiments about 60 seconds

Do Click Experiments. Start with Evaluation = Unseen stations · future test period. Point to RMSE and SD. Switch to Known stations · future test period. Scroll to Sensor corruption sweep.

Say "Unseen-location mapping and known-station forecasting are different tasks. LSTM uses a target's own history, so it has no unseen-station mapping score. The corruption sweep tests how errors change as more clients have severe simulated faults."

Demo part two the Simulation lab

Target time: 3 minutes. Click Simulation lab in the sidebar. This page contains saved extended experiments and a separately computed interactive sensor scenario. Explain that distinction before changing controls.

5 Show the extended evaluation

Do Point to Five-fold leave-stations-out evaluation, then Early input calibration ablation. Read the trust/FedAvg CV RMSEs and the corrected trust spatial RMSE.

Say "Every retained location is held out once per seed across five folds. Trust CV RMSE is 96.69 and FedAvg is 96.16. In the separate fixed-holdout calibration ablation, corrected trust is 78.67. These tables use different evaluation setups, so we keep them separate."

Do Briefly point to Relative-error and temporal reputation sensitivity. Say that Kanpur relative errors anchor a separate assumed sensor scenario; temporal reputation smooths update agreement over rounds. This table does not establish a field-validated hardware model.

6 Run three controlled sensor cases

Scroll to Sensor scenario. Set Station to CRRI Mathura Road, Delhi - IMD and keep it unchanged for all three cases. Enter the values for one case (column), then click Run scenario. Repeat for B and C. Dropout fraction 0.05 means 5%, not 0.05%.

Control	A Ideal	B Default errors	C Stronger drift
Bias µg/m³	0	20	20
Gain	1	1.1	1.1
Annual drift µg/m³	0	20	80
Noise SD µg/m³	0	10	10
Dropout fraction	0	0.05	0.05
Press	Run scenario	Run scenario	Run scenario
Expected raw → corrected RMSE	0.0 → 0.0	59.4 → 18.1	114.0 → 64.1

Expected values are rounded µg/m³, verified for this station with fixed seed 17. Case A uses 1,262 valid test pairs; B/C use 1,213. Different stations or controls give different outputs. The first 10% of paired hours fit the correction; only valid future-period pairs score it.

Say "Case A is a sanity check: an ideal sensor matches its reference. Case B adds offset, gain, noise, drift and missing readings; early calibration reduces future error. Case C increases annual drift to 80: the old correction becomes less reliable, and corrected RMSE rises. This illustrates why calibration can need refreshing."

Point to the three plotted series: reference, simulated and calibrated. The plotted curves are monthly means; RMSE uses valid hourly test pairs. These RMSEs assess one virtual sensor and its correction, not the graph model. Run scenario does not retrain or update the global FL checkpoint.

7 Show the added real pollutant data

Do Scroll to Expanded observations. Set Pollutant to PM10, then NO_2; the chart updates automatically. Scroll briefly to Station coverage audit.

Say "These are observed monthly trends, with 323,061 PM10 and 325,379 NO_2 source rows. Coverage varies by station and month. The audit exposes missing hours, coverage and concentration summaries. These pollutants expand context, not PM2.5 training labels."

Demo part three evidence and closing

Target time: 2–3 minutes. Finish with what the evaluator can verify: a specified privacy test, documented data and reproducible files.

8 Privacy probe

Do Click Privacy probe. Point to graph/trust AUC and the threat-model explanation.

Say "The honest-but-curious server sees individual client updates. Our loss-improvement probe asks whether an update reduces a candidate training window's loss more than a reserved nonmember's loss. We measure AUC and attack advantage. Near-chance AUC for this limited attack does not prove privacy. Windows may overlap; we have neither secure aggregation nor differential privacy."

9 Datasets and exports

Do Click Datasets & exports. Point to source names and licenses. Scroll to Submission exports. Click Dataset manifest, then Datasets used ZIP if ma'am asks for the actual files.

Say "We retain source observations, cleaned and simulated derivatives, source roles, CC BY attribution and SHA-256 integrity checks. The manifest is an audit of files; it is not a model-accuracy score. We submit the actual dataset package separately from the source-code package."

The visible export links are Datasets used ZIP, Complete project ZIP, Implementation report, Held-out predictions, Dataset manifest and Comparison table. "Implementation report" opens the original fixed-holdout methods/results Markdown document; extended findings and the final PDF are linked from GitHub's README. "Held-out predictions" is a CSV.GZ download, not an on-screen chart.

10 Open the repository and close

Do Open https://github.com/rizzit17/airtrust. Show README → Final report PDF, 25-paper literature comparison and Requirements and teacher-feedback evidence. If asked about execution, show the successful Actions run or run python -m pytest -q in a second terminal.

Say "Our source, saved models, station-fold audits, results and data snapshot are reproducible. The test suite currently has 17 passing checks. Our conclusion is measured: input calibration helps this simulation, while trust weighting alone does not consistently improve on FedAvg. Field co-location, seasonal transfer and stronger privacy protection are the next steps."

If something does not load

Browser cannot connect: check the server terminal; run start.bat from airtrust. Port already occupied: use the existing app, or start python -m uvicorn airtrust.server:app --host 127.0.0.1 --port 8001 and open that port. Avoid launching two servers on port 8000.

Missing data on a fresh clone: run python scripts/prepare_local_data.py. Download ZIP returns unavailable: run python scripts/build_submission.py, then python scripts/package_submission.py before the demonstration. Map seems unchanged: press Update map after changing its selector or slider. Keep the terminal open throughout.

Two minute fallback if time is cut short

Overview and opening (20 s) → Pollution map, press Update map (30 s) → Simulation lab, show CV/calibration and run default case B (50 s) → state the honest conclusion and show dataset ZIP/GitHub (20 s).

Viva answers and final revision

What is our actual contribution?

An end-to-end, calibration-aware FL mapping experiment with reproducible spatial holdouts, robust comparisons, error sensitivity and an interactive demonstration. We tested the weighting idea rather than assuming it must win.

Why federated learning if the data are already public?

Public data let us simulate and evaluate a future distributed-owner setting. Our aggregator accepts model parameters, but the single-machine evaluator loads the public observations. We claim algorithmic data locality, not a protected deployed network.

How did we avoid spatial leakage?

Each held-out station is excluded from training labels, validation, calibration and donor histories. The graph model sees other allowed stations' past readings plus coordinates; held-out truth is used only for final evaluation.

Why can trust weighting lose?

It changes the influence of model updates without correcting biased donor inputs. Calibration can become stale, and reducing a client's weight can reduce geographic diversity. Our three-seed primary test does not show a significant trust gain.

Why not simply call the LSTM our best mapper?

The LSTM uses a known target's own history. A completely unseen target has no permitted history in our mapping evaluation. Comparing its RMSE directly with the graph mapper would mix information sets.

Is this live training or live data?

The dashboard loads previously trained checkpoints and measured tables. Update map performs inference; Run scenario generates a deterministic sensor diagnostic and fits an early linear correction. Observations are historical, not a live feed.

What is calibration versus trust?

Calibration changes a sensor's values using paired reference observations. Trust changes a client's aggregation weight based on calibration residuals. Our separate ablation shows why those operations must be distinguished.

Why Beijing and why the extra pollutants?

Beijing is a separate reference-grade persistence check. Delhi PM10 and NO_2 support contextual trends. Our reported FL target remains Delhi PM2.5; no cross-city FL or multi-pollutant model claim is made.

Does FL guarantee privacy or lower communication?

No. Updates can leak information. We report one named attack and logical payload counts, without cryptographic privacy, radio measurements or a guaranteed bandwidth advantage.

What should we disclose about readiness and AI?

The software and evaluation are implemented; publication has not been submitted and field hardware validation remains pending. AI assisted implementation, documentation and debugging; the team must understand and verify the work.

Last facts to revise

36 retained sites • 300,355 valid PM2.5 observations • 8 clients • seeds 17/29/43 • 20 rounds • 5 spatial folds • CV RMSE trust 96.69 versus FedAvg 96.16 • corrected fixed-holdout trust 78.67 µg/m³ • 17 automated checks.